

Element E: Application of STEM Principles and Practices

The Makey Makey project exemplifies applying STEM principles and practices by integrating science, technology, engineering, and mathematics into a hands-on learning experience. In



science, students explored the conductivity of materials, using items like fruits and aluminum foil to create interactive inputs that respond to touch, while also understanding the principles of circuits and grounding through conductive tape. The technology aspect was highlighted by utilizing the Makey Makey board to connect these physical components to sound output, allowing for an engaging musical experience. Engineering principles came into play as students designed their projects, considering factors like layout and accessibility to ensure they were

user-friendly. Finally, mathematics was involved in measuring and arranging materials accurately to create functional instruments. This project not only reinforced key STEM concepts but also encouraged collaboration and problem-solving as students worked together to build and test their innovative creations.